

CivicRoute Service Request Triage

Interpretable NLP, harder holdout evaluation, and manual review

Project type	Applied NLP
Source records	175,784
Prepared rows	36,805
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1. Executive summary

CivicRoute prepares 175,784 source records into 36,805 labelled and deduplicated service requests across eight queues.

A standard random holdout reaches 0.995 macro F1. A harder split that excludes complete category codes from training reaches 0.974 macro F1 on 12,789 requests.

At the 0.58 threshold, the harder split routes 84.8% automatically and accepted suggestions reach 99.9% accuracy in this snapshot.

2. Problem and purpose

Citizens describe service problems in free text while service teams organise work into structured queues. That mismatch can create avoidable transfers and slow the first response.

The project asks whether a compact, interpretable model can suggest a useful first queue without hiding uncertainty.

The output is a triage aid. It does not file a complaint, validate facts, determine priority or legal rights, or replace an official portal.

3. Source and privacy boundary

The source is the Government of India complaint corpus published on Kaggle as a grievance report. The dataset page states an MIT licence and provides a category mapping workbook.

Grievance is the source system's formal term for a complaint or service request. CivicRoute uses clearer service-request language in its interface while preserving the original term in provenance notes.

The source file is named `no_pii_grievance.json`, but complaint text can still contain details entered by a citizen. Prepared text is therefore not published in Git or exposed as a searchable corpus.

4. Preparation pipeline

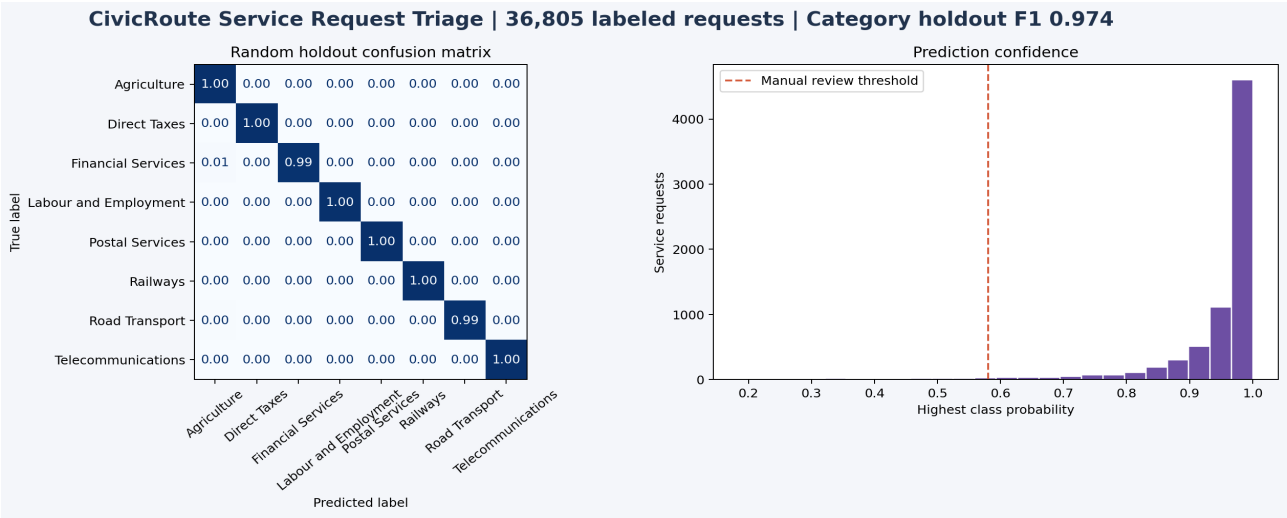
The acquisition script maps category codes to organisations, selects eight well-represented service queues, removes insufficient text, cleans repeated redaction markers, and drops exact duplicate complaint text. 38,779 rows pass selection before deduplication.

The model table keeps a project identifier, request text, queue, organisation code, category code, and category description. Generated raw and prepared text files stay outside Git.

Queue counts are imbalanced, so the classifier uses class weights and evaluation reports macro F1 alongside accuracy.

Service queue	Prepared rows
Labour and Employment	9,458
Financial Services	6,272
Direct Taxes	5,978
Agriculture	5,220
Telecommunications	3,622
Postal Services	3,058
Railways	2,048
Road Transport	1,149

5. Evaluation dashboard



6. Language model

TF-IDF represents unigrams and bigrams while reducing the influence of generic repeated words. The vocabulary is capped so the complete workflow remains practical on a student laptop.

Class-balanced logistic regression returns one probability per service queue and allows influential terms to be inspected. Text preparation and classification are stored as one reproducible pipeline.

This design is smaller, cheaper, and easier to explain than using a general-purpose language model for a fixed eight-queue task.

7. Confidence and manual review

Requests below 0.58 confidence are not routed automatically. The best suggestion remains visible as supporting information for a reviewer.

Messages shorter than 25 characters are sent to review before model inference. Multi-topic, emergency, personally identifying, and unfamiliar requests should also be reviewed.

A confidence threshold cannot detect every confidently wrong result. It must be recalibrated when queues, vocabulary, or source systems change.

Harder holdout measure	Result
Automatic coverage	84.8%
Accepted-route accuracy	99.9%
Unseen category codes	90

8. Evaluation interpretation

The random split is useful for comparison but can be optimistic when training and test rows share administrative category language.

The category-group holdout keeps complete category codes out of training. Its lower score is a more credible measure of how the model handles less familiar request types and is therefore the headline result.

The corpus is still English-dominant and historical. Future evaluation should include time-based splits, near-duplicate detection, independently written samples, and multilingual requests.

9. Governance, reproducibility, and next steps

The repository documents source, licence, privacy boundary, included queues, both evaluation splits, threshold, limitations, and the official CPGRAMS link.

Acquisition, preparation, training, inference, evidence generation, tests, and this report are versioned. The README provides the full rebuild sequence.

Next steps are queue-specific thresholds, drift monitoring, multilingual evaluation, a reviewer interface, emergency-message detection, and strict retention controls for request text.

Risk	Control
Wrong automatic queue	Confidence threshold and human review
Overstated evaluation	Category-group holdout as headline result
Privacy exposure	Generated request text excluded from Git
Taxonomy drift	Version mappings and reevaluate